General purpose PLL

2026 May 27

Super Long Wave

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Jay

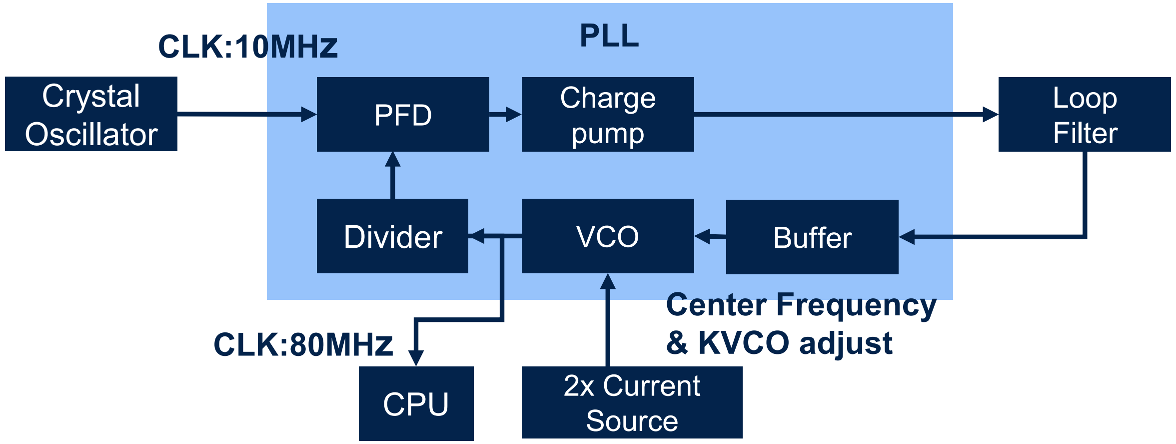
Teddy

Butter cutter

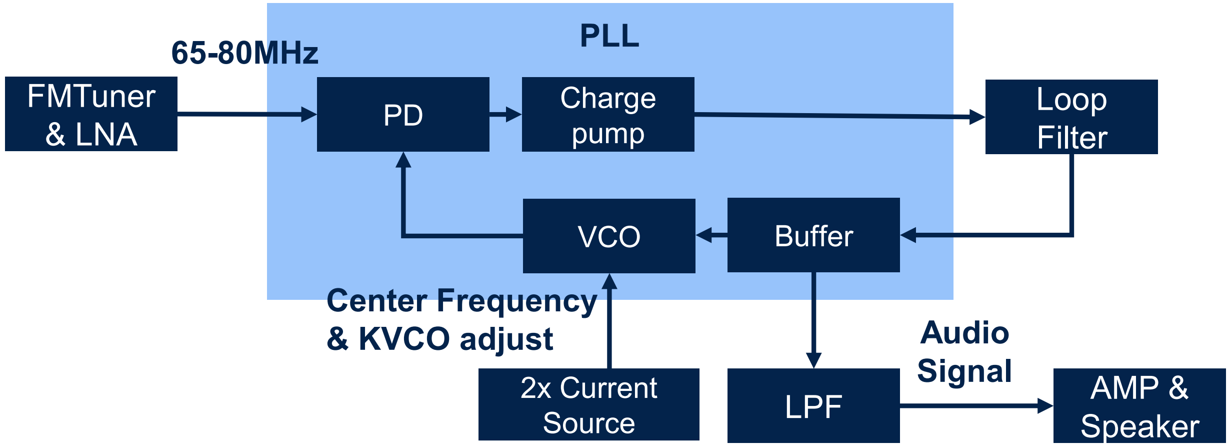
Agenda

Project information

Clock multiplier mode



FM demodulator mode



- Goal
  - Creating general purpose PLL
  - Creating a part of FM Radio component
- Design
  - PLL has two mode
  - Clock multiplier mode
  - FM demodulator mode
- Application
  - Digital clock generator & FM Radio
- Reference
  - TI CD4046B Application Report SCHA002A

Team background

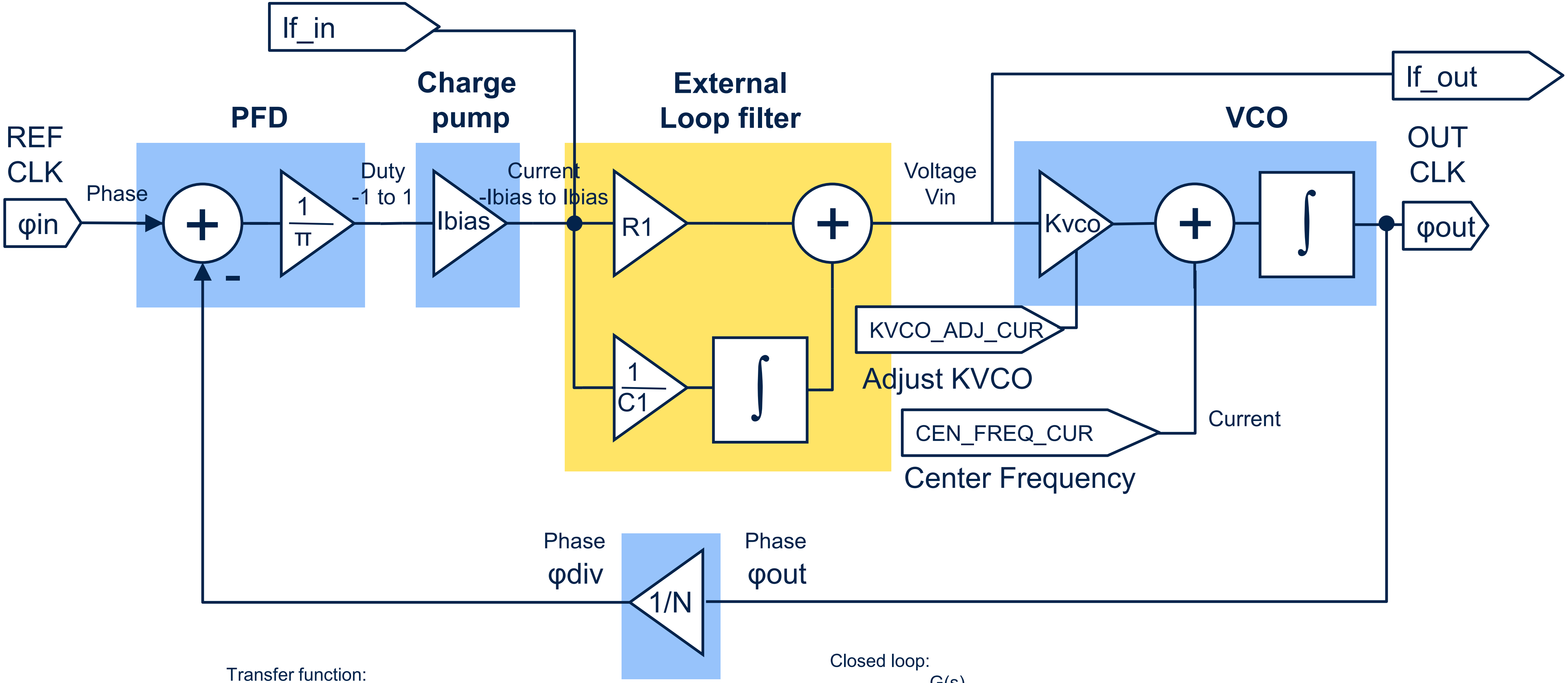
	Yutaka KOTANI	Jay	Teddy	Butter Cutter
Academic experience	Bachelor of Computer Science			
Work experience	Field Application engineer in Semiconductor company			

Specification

Target Spec	Min	Typ	Max
Pin count		11	
Estimated Area		999um2	
Vdd voltage [V]	3	3.3	3.6
pll_in voltage [V]	0.8	-	2.5
pll_out voltage [V]	0		vdd
Clock multiplier mode	**Min**	**Typ**	**Max**
pll_in frequency [MHz]	9.9	10	10.1
pll_out frequency [MHz]	79.9	80	80.1
Divider ratio		1/8	
Cen_freq_cur [uA]	90	100	110
FM demodulator mode	**Min**	**Typ**	**Max**
pll_in frequency [MHz]	76	-	90
dmod_out voltage [V]	0	-	vdd
Cen_freq_cur [uA]	10	-	110

Pin	Pin name	Notes
1	VDD	3.3V
2	GND	
3	PLL_IN	Reference clk input
4	PLL_OUT	Clock multiplier output
5	CEN_FREQ_CUR	Center frequency bias current
6	KVCO_ADJ_CUR	kvco adjust bias current
7	MODE_SEL	Hi: Use FM demodulator Low: Use Clock multiplier
8	EXT_FIL_OUT	To Band pass filter 10.7MHz
9	EXT_FIL_IN	From Band pass filter 10.7MHz
10	LF_IN	External Loop filter input
11	LF_OUT	External Loop filter output

Block diagram - Clock multiplier mode



Divider

Transfer function:

$$G(s) = \frac{Ibias}{\pi} * (R1+1/C1s) * \frac{Kvco}{s}$$

$$H(s) = 1 / N$$

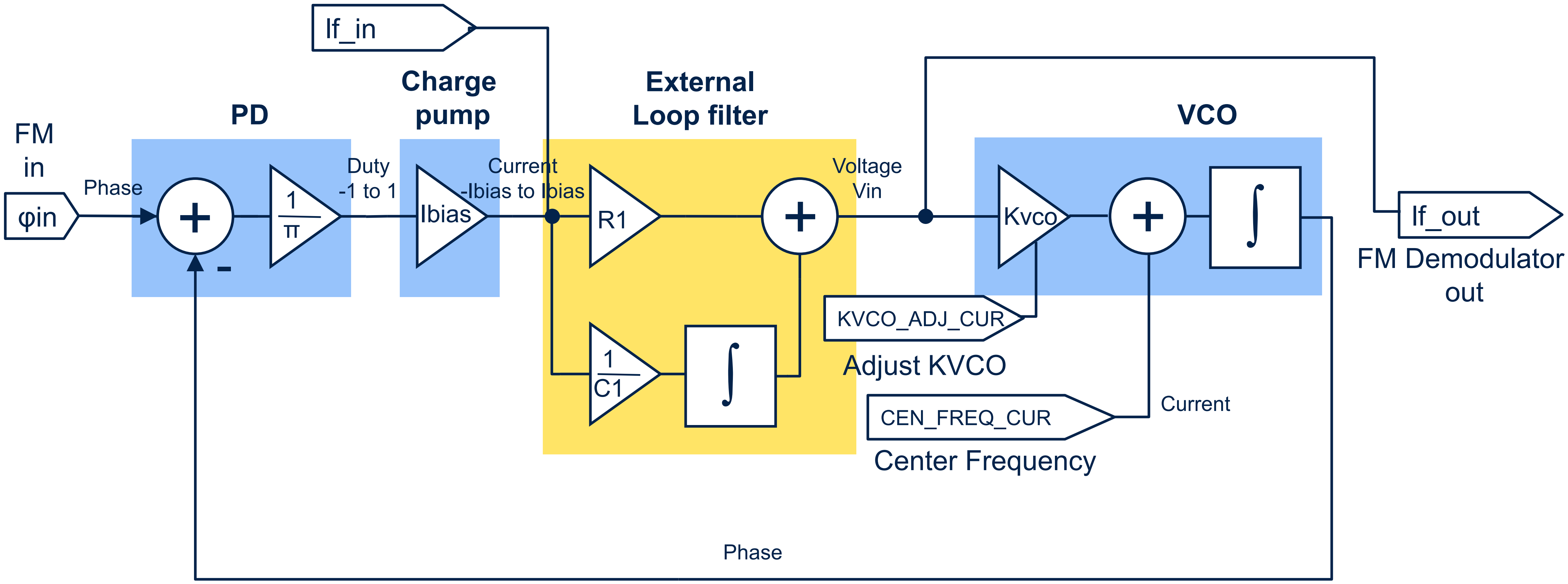
Closed loop:

$$W(s) = \frac{G(s)}{1 + G(s)\,H(s)}$$

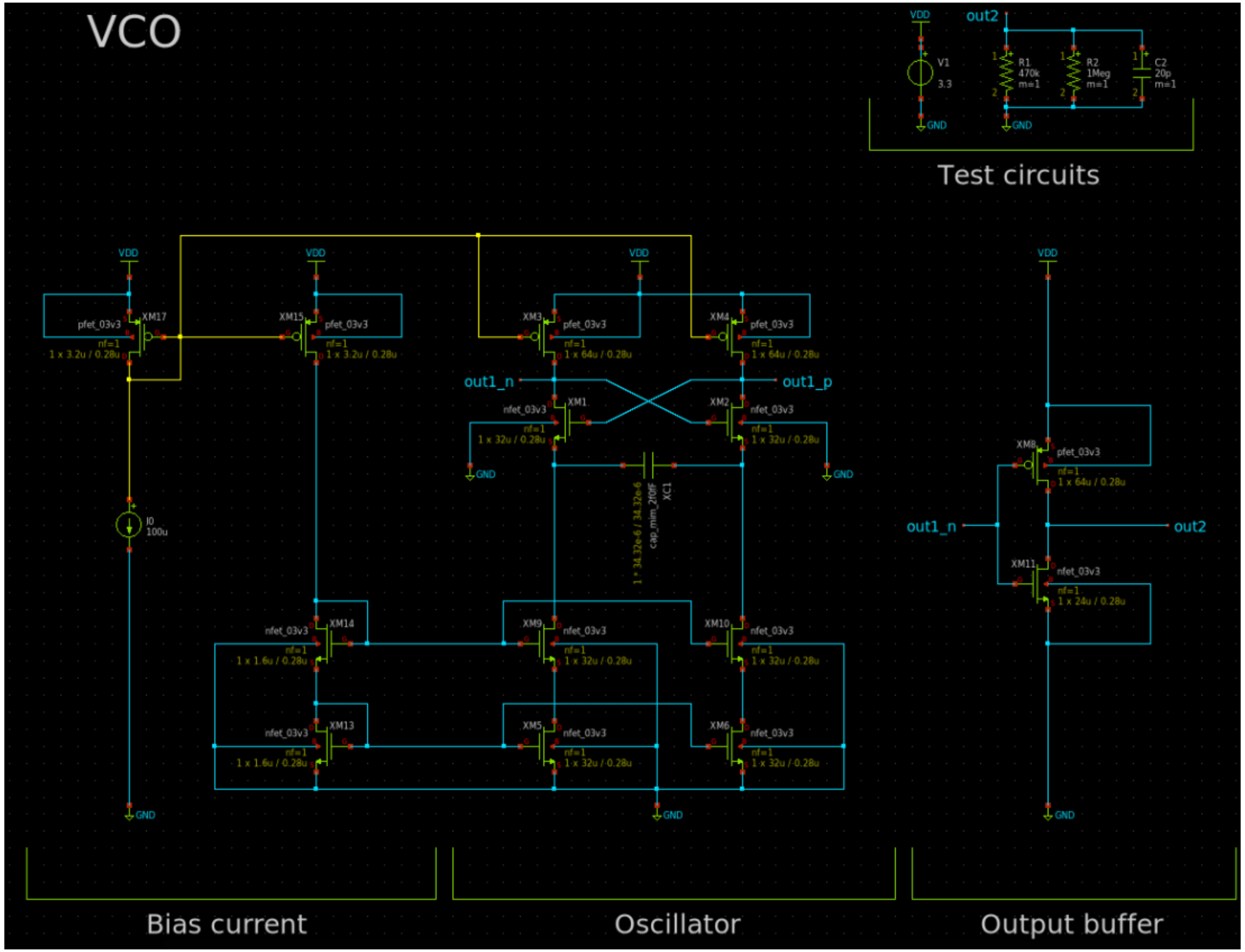
Open loop:

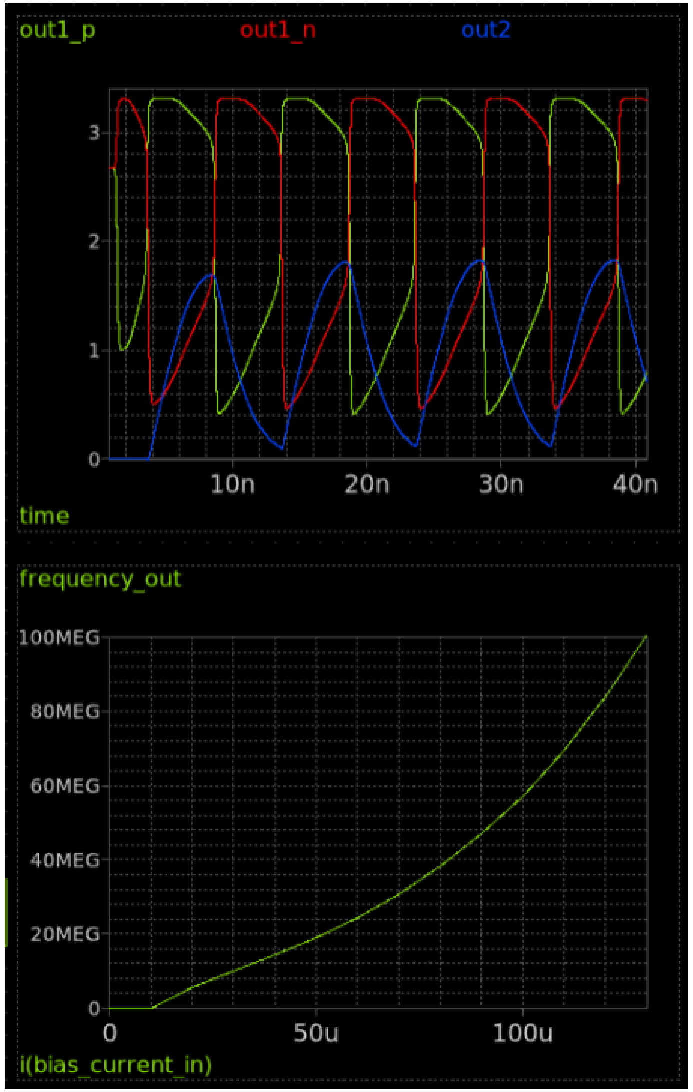
$G(s)\,H(s)$

Block diagram – FM demodulator mode

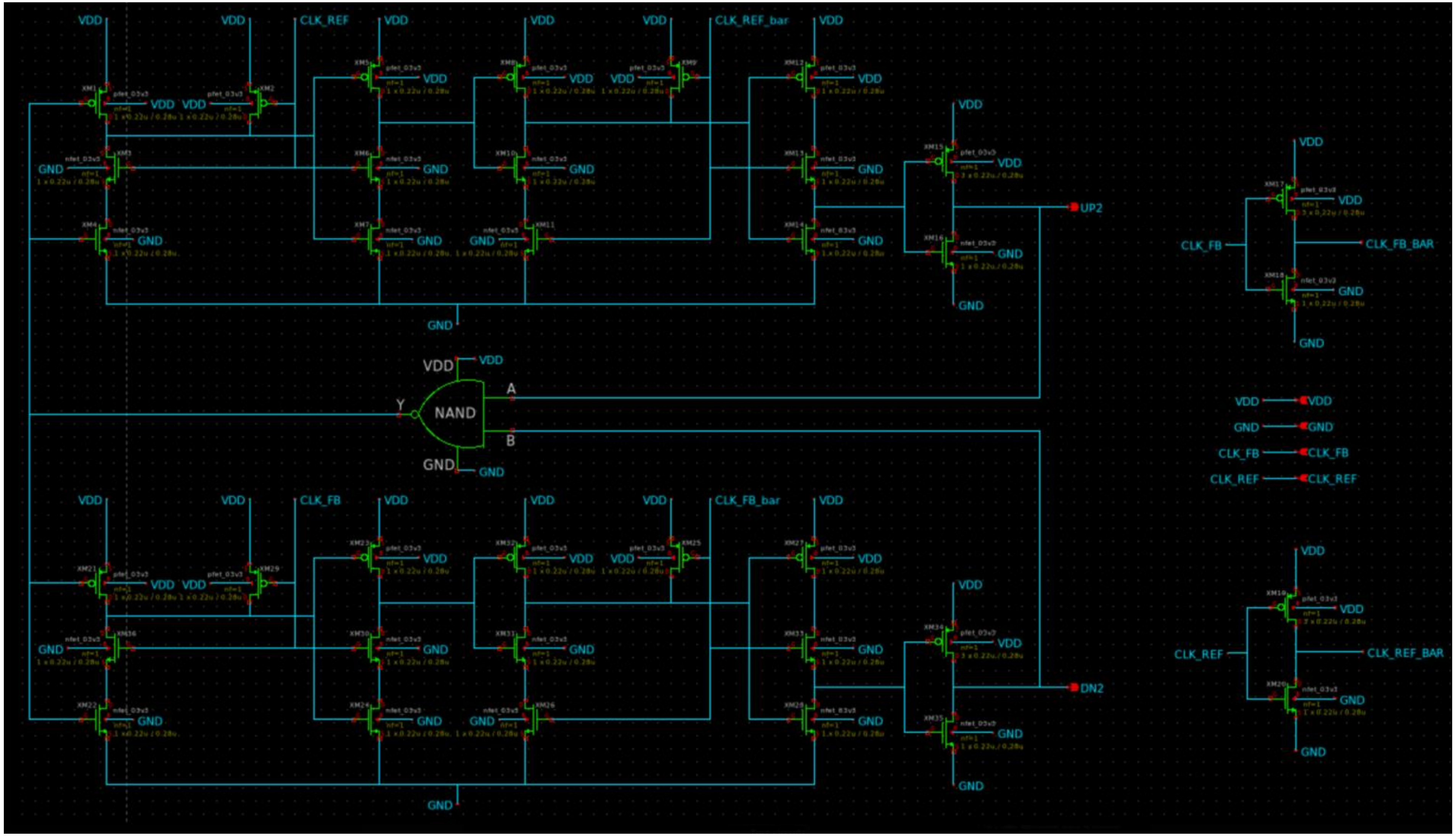


Circuit design - VCO

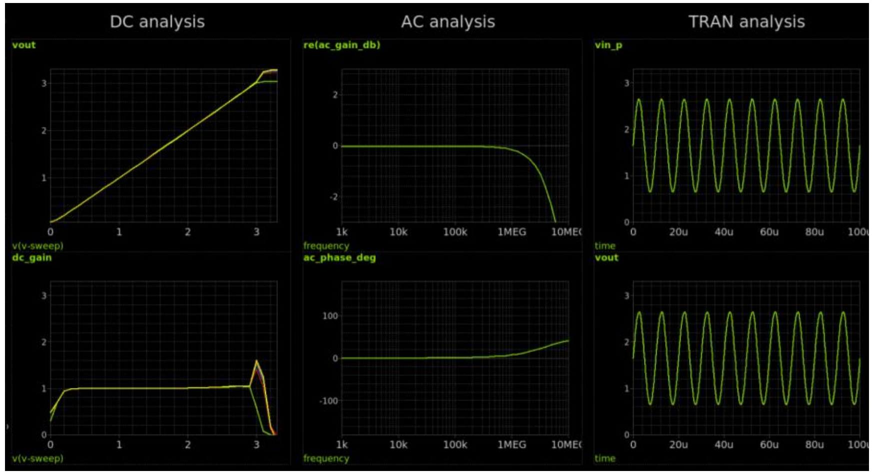


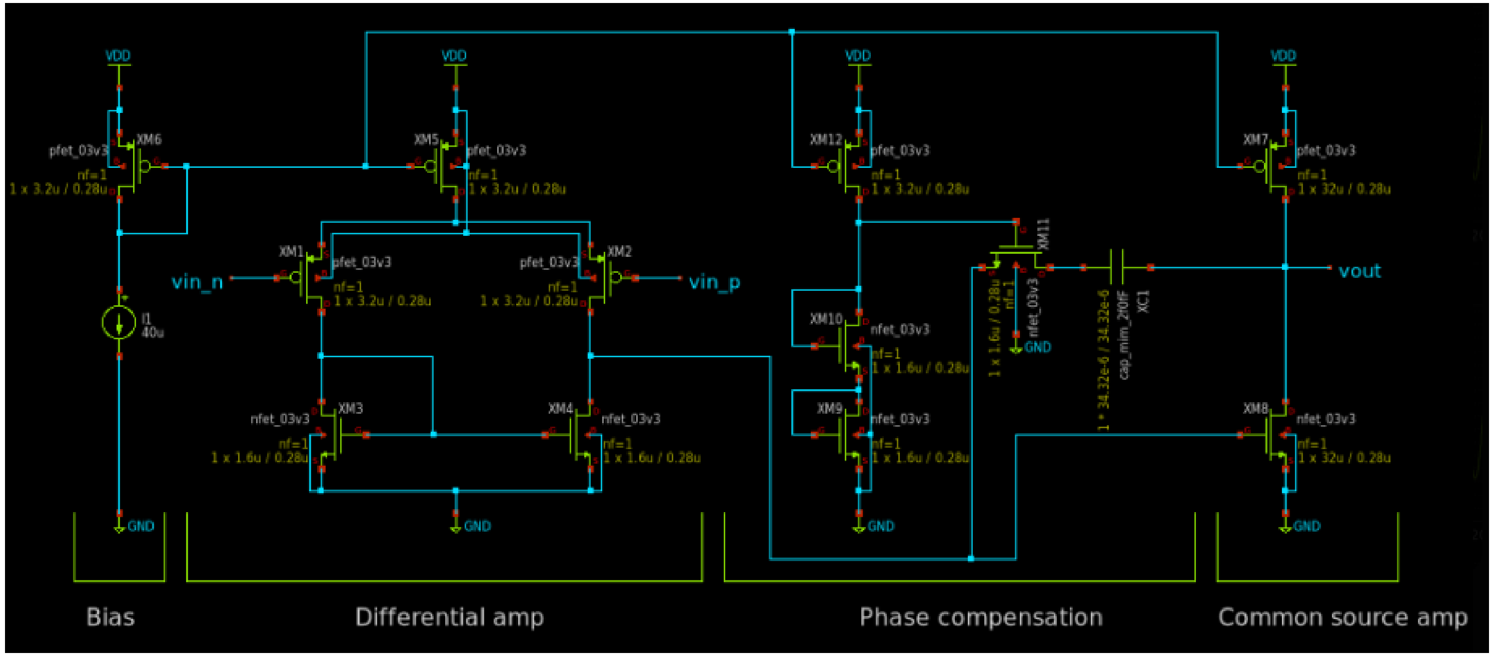


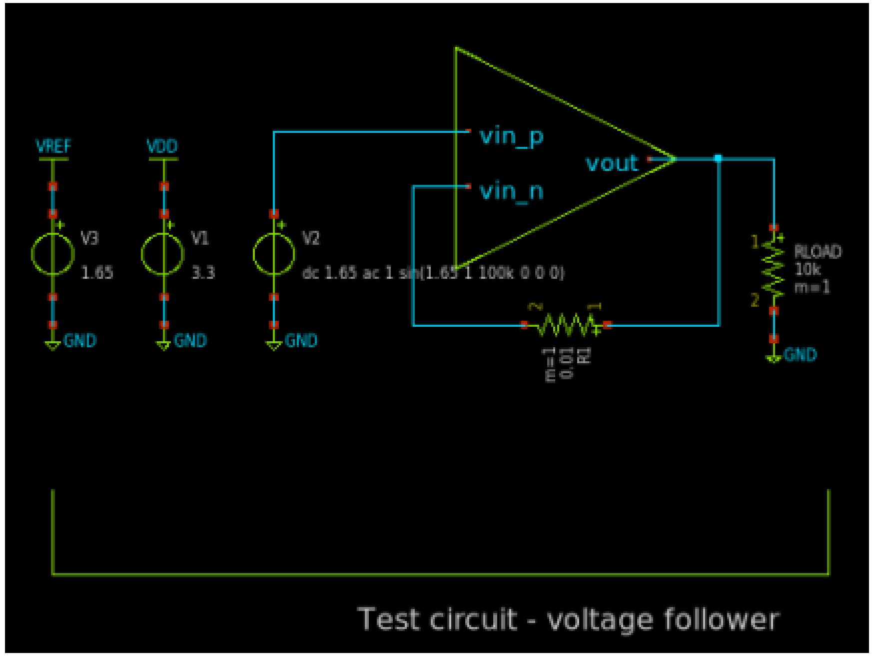
Circuit design – PFD



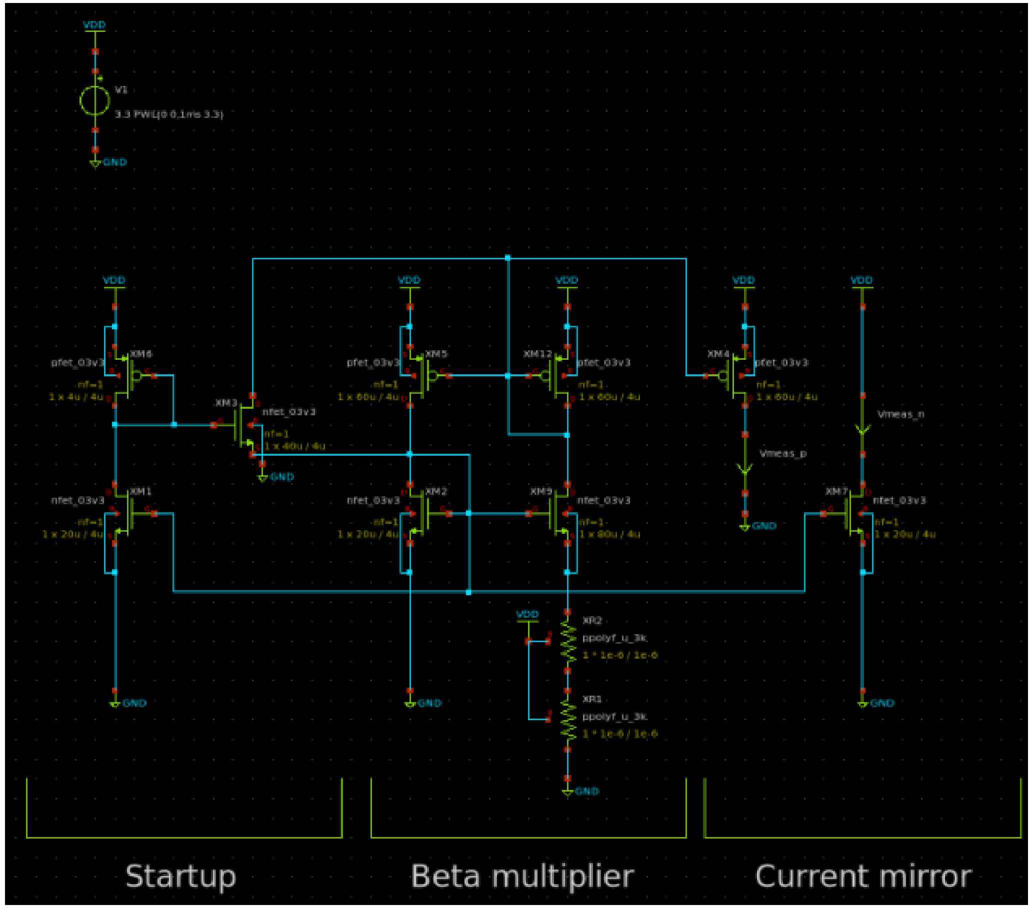
Circuit design - OPAMP

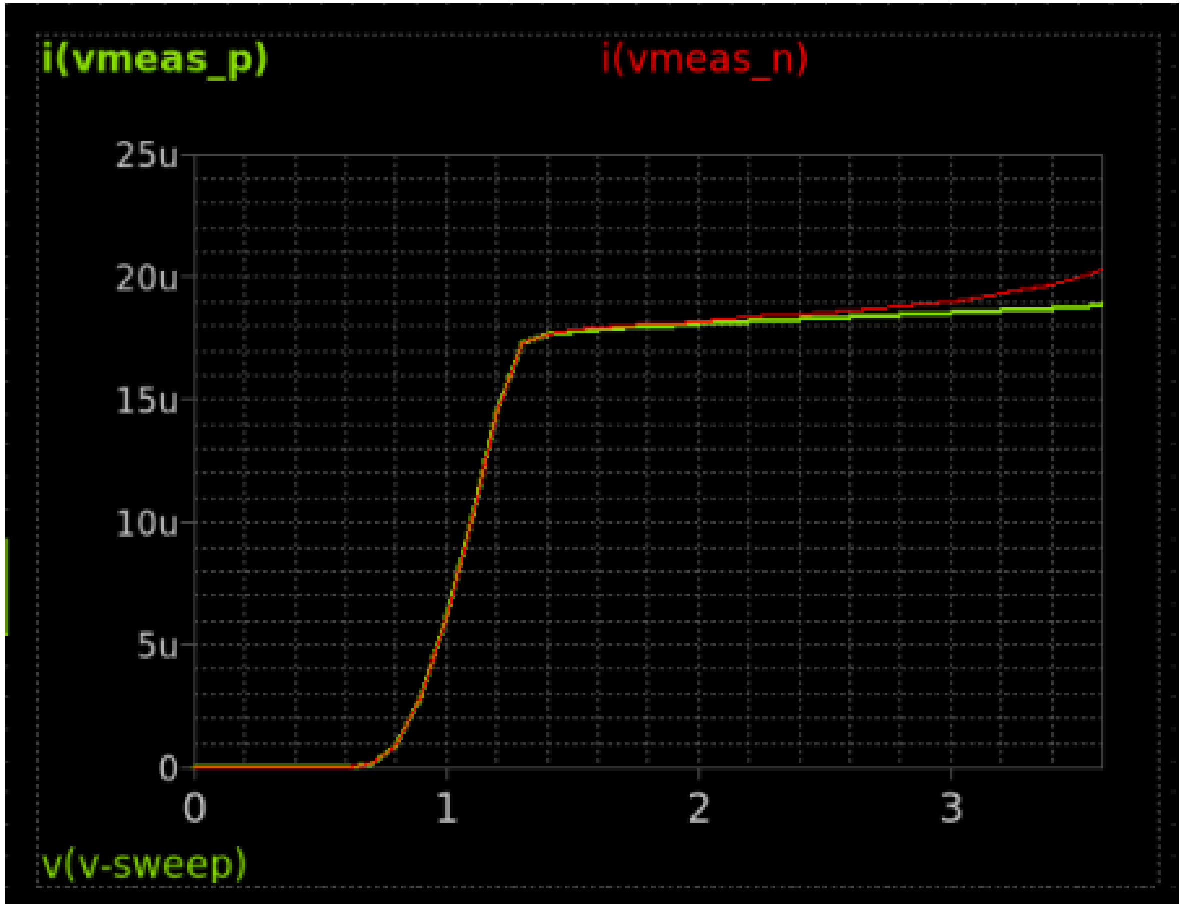






Circuit design – Beta multiplier reference





Layout (T.B.D)

Evaluation (T.B.D)

- Put evaluation result here
- Insert graph to left

Schedule

Week	Date	Event	Track Details
Week 23	June 5	Team Formation Deadline	
Week 24	June 12	Project Proposal Review	
Week 27	July 3	Schematic Review	
Week 28	July 10	Simulation Review (blocks)	
Week 29	July 17	Simulation Review (top level)	
		READiness Check & Go/No-go Decision	
Week 33	Aug 14	Layout Review (blocks)	
Week 34	Aug 21	Layout Review (top level)	
Week 35	Aug. 28	Verification	
		Final Chip Review	
		Final Submission	
		Upload of Chip Test Results	